Day 25 | Monday

25

December 14, 2026

Review: numbers, units and formulas

9:00 AM–10:15 AM • Math Review • 75 min

Review: geometry and job problems

10:30 AM–12:00 PM • Math Review • 90 min

Week 8 • 2 scheduled blocks • 165 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

9:00 AM–10:15 AM • 75 min

Review: numbers, units and formulas

9:00 AM–9:05 AM	**Start**	5
9:05 AM–9:10 AM	**Recall**	5
9:10 AM–9:25 AM	**Teach**	15
9:25 AM–9:40 AM	**Together**	15
9:40 AM–10:00 AM	**Practise**	20
10:00 AM–10:10 AM	**Check**	10
10:10 AM–10:15 AM	**Wrap up**	5

Math Review • Math Week 8, Lesson 1

10:30 AM–12:00 PM • 90 min

Review: geometry and job problems

10:30 AM–10:35 AM	**Start**	5
10:35 AM–10:40 AM	**Recall**	5
10:40 AM–10:55 AM	**Teach**	15
10:55 AM–11:10 AM	**Together**	15
11:10 AM–11:45 AM	**Practise**	35
11:45 AM–11:55 AM	**Check**	10
11:55 AM–12:00 PM	**Wrap up**	5

Math Review • Math Week 8, Lesson 2

Break: 10:15 AM–10:30 AM • 15 minutes • excluded from teaching time

Review: numbers, units and formulas

Today’s goal: Retrieve core methods across the course and explain when each method is appropriate.

Identify one target skill and the checking habit to use today.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Stations: 1B Q3 p.8; 4A Q14(a),15(a) p.26; 5A Q12–13 p.33; 6A Q19 p.40; 7A Q11–12 p.45; 8C Q1–2 p.55.

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Recall and reconnect

Use learners' two priorities from Week 7. Ask each to identify a useful checking strategy.

Review previous feedback and attempt the retrieval prompt without displayed solutions.

Resources: Stations: 1B Q3 p.8; 4A Q14(a),15(a) p.26; 5A Q12–13 p.33; 6A Q19 p.40; 7A Q11–12 p.45; 8C Q1–2 p.55.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Choose the method before the keys

1. At the number station, explain the operation choice and any rounding needed for a whole purchase unit.

2. At the fraction/decimal station, show the regrouping or conversion that keeps the quantity unchanged.

3. At the conversion/scale station, record the factor, its direction and both units.

4. At the formulas station, find the missing dimension and substitute it back into the original relationship.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Let your error log guide the review

1. Begin with the two revision priorities from Week 7 and select a checking strategy for each.

2. Look for the step behind the error: regrouping, a wrong factor, a wrong denominator or missing brackets.

3. Use the selected Textbook chapters, formula sheet p.219 and conversion factors pp.220–221 to explain why the method applies.

4. Keep written working and record whether the solution was independent, tool-checked or completed with a prompt.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Rotate through four method stations

1 • NUMBER DECISIONS
1B Q3 p.8.
Explain operation choice and whole-purchase-unit rounding.

2 • FRACTIONS / DECIMALS
4A Q14(a),15(a) p.26;
5A Q12–13 p.33.
Explain regrouping or conversion.

3 • CONVERSION / SCALE
6A Q19 p.40;
7A Q11–12 p.45.
Show factor, direction and units.

4 • FORMULAS
8C Q1–2 p.55.
Find the missing dimension, then substitute it back.

Instructor: model one error from the logs. Partners explain one method at each stop.
Tools after paper: Calculator, Unit Conversions, or a Shape Canvas missing-dimension rectangle.

Exercise Book practice stations • No displayed answers during first attempts

Textbook | page 127

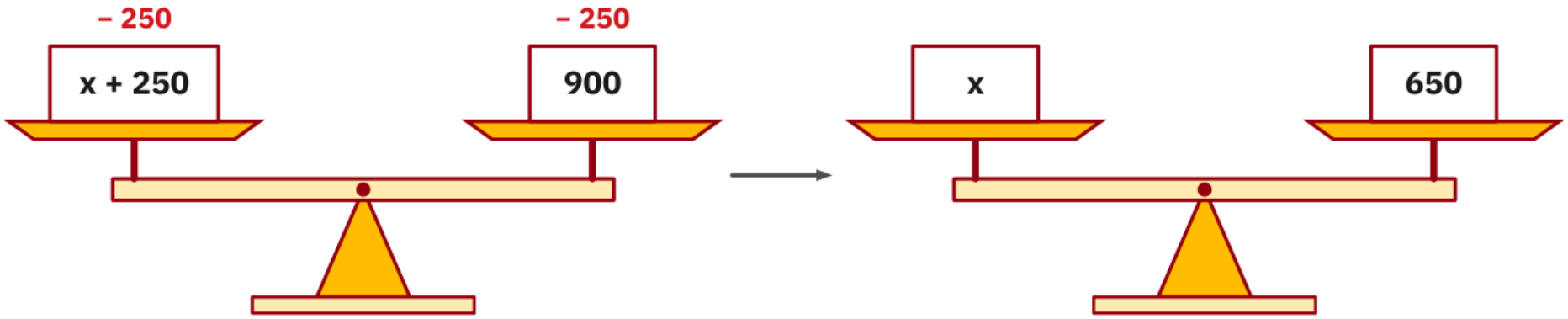


Figure 8.1 Take 250 off both pans and the scale stays level. The unknown x is left on its own.

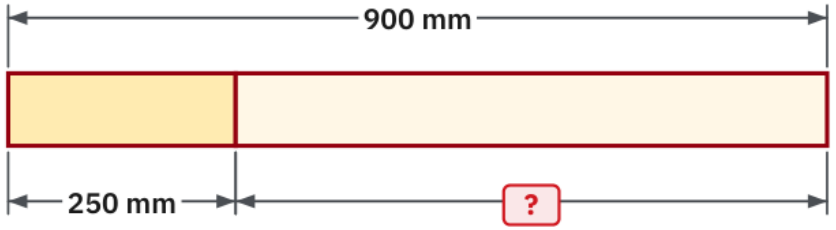
Figure 8.1: inverse operations preserve equality.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Textbook | page 127

Example 8.4 Undoing an operation

A 900 mm bar is cut into a 250 mm piece and a second piece, so $x + 250 = 900$. Solve it. Then solve $4 \times x = 1{,}000$ and $x \div 3 = 125$.



SOLUTION

1. 250 was added to x. Subtract 250 from both sides: $x = 900 - 250 = 650$. Check: $650 + 250 = 900$.
2. x was multiplied by 4. Divide both sides by 4: $x = 1{,}000 \div 4 = 250$. Check: $4 \times 250 = 1{,}000$.
3. x was divided by 3. Multiply both sides by 3: $x = 125 \times 3 = 375$. Check: $375 \div 3 = 125$.

Answer **The second piece is 650 mm. Then x = 250 and x = 375.**

Example 8.4: name the operation, undo it, then substitute back.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Spot the mistake. Explain the correction.

A conversion response says “multiply by the factor” but omits the starting and target units. What still needs to be justified?

Discuss the method before checking the answer.

Name both units and show why the chosen factor and direction produce the requested unit. Check the result with Unit Conversions after the paper attempt; retain the factor and working.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Work together

Pairs rotate through number, fraction/decimal, conversion and formula stations; explain one solution aloud at each stop.

Explain the method and check directions; work with a partner only where permitted.

Resources: Stations: 1B Q3 p.8; 4A Q14(a),15(a) p.26; 5A Q12–13 p.33; 6A Q19 p.40; 7A Q11–12 p.45; 8C Q1–2 p.55.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Your scheduled practice

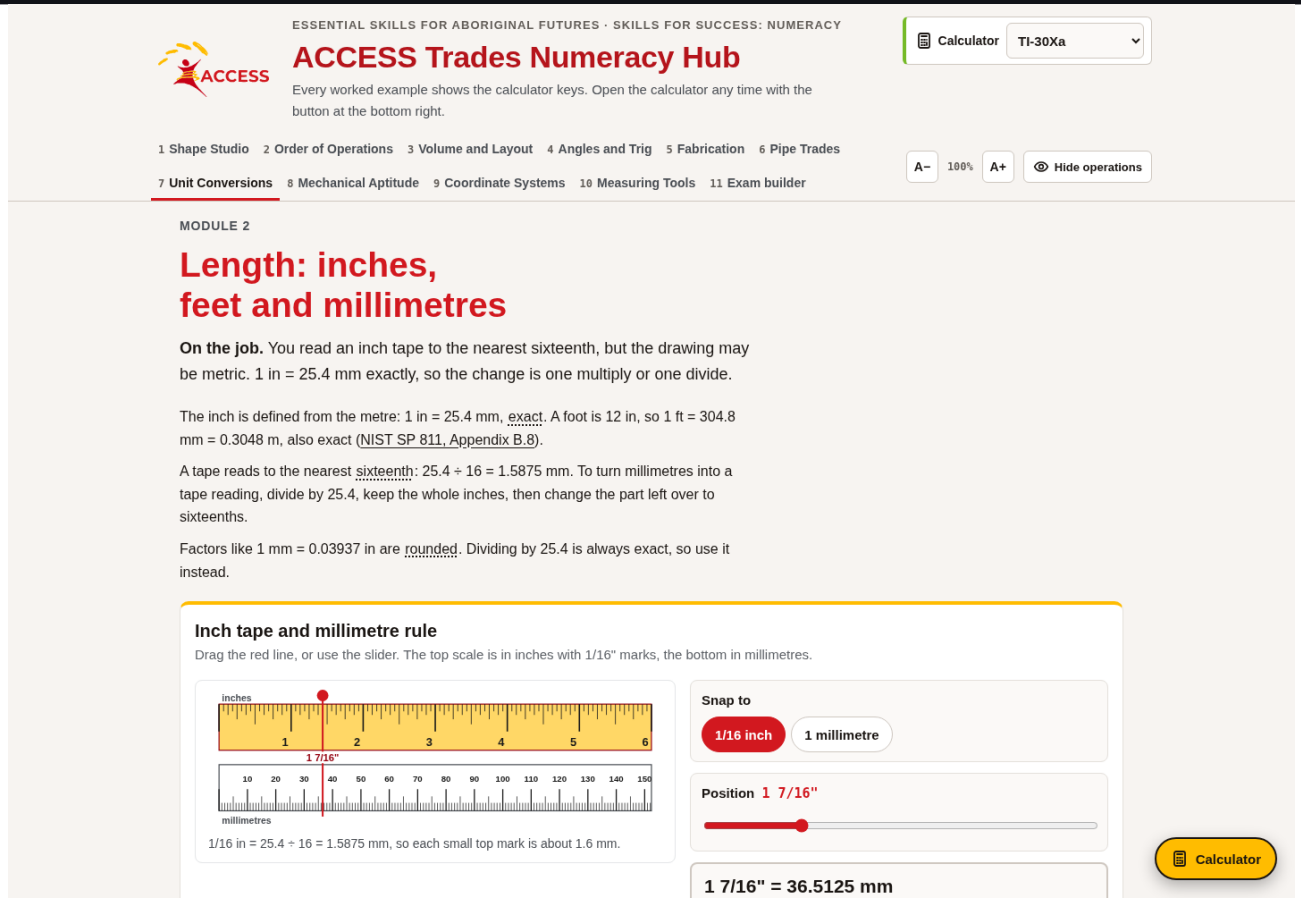
Complete the listed station questions; start with the two highest-need topics. Keep written working and note whether a tool or prompt was required.

Keep numbered written solutions and units. Use tools only under the conditions stated for this block.

Resources: Stations: 1B Q3 p.8; 4A Q14(a),15(a) p.26; 5A Q12–13 p.33; 6A Q19 p.40; 7A Q11–12 p.45; 8C Q1–2 p.55.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Open Numeracy Hub



Navigation example. Use the values and units in the assigned task.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Practise in Numeracy Hub and Shape Canvas

1. At the relevant stations, write the selected book solution first and state the method and units.

2. Use Numeracy Hub Calculator for number checks or Unit Conversions for length practice; at the formulas station, use the scheduled Shape Canvas rectangle with one missing dimension.

3. Compare the tool check with your working, and substitute the missing dimension back where relevant. Mark whether each solution was independent, tool-checked or prompted.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Practise from the Exercise Book | 1B Q3

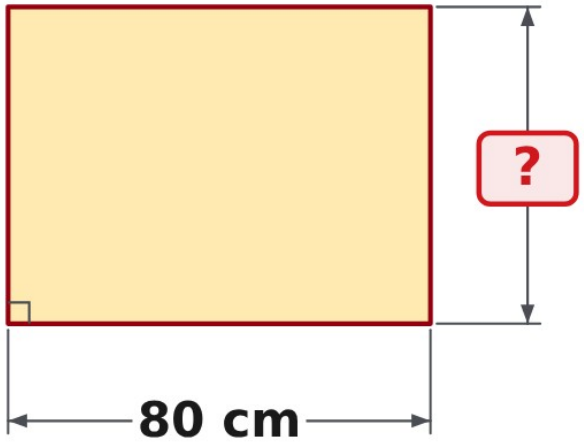
A millwright sets 14 baseplates. Each baseplate takes 8 anchor bolts. Anchor bolts come in boxes of 25. How many bolts are needed, and how many boxes?

Write the operation, working and units before checking.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Practise from the Exercise Book | 8C Q1

The rectangle has an area of 4,800 cm². Find the width.



Write the operation, working and units before checking.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Check independently

Exercise Book 6A Q19, 7A Q11 and 8C Q2; identify the appropriate method for each.

Submit work and demonstrate the checking strategy; identify a remaining difficulty.

Resources: Stations: 1B Q3 p.8; 4A Q14(a),15(a) p.26; 5A Q12–13 p.33; 6A Q19 p.40; 7A Q11–12 p.45; 8C Q1–2 p.55.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Independent check | 6A Q19

At a scale of 1 : 40, a line measures 85 mm. How long is the real length?

Answer independently. Keep worked solutions closed.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Independent check | 7A Q11

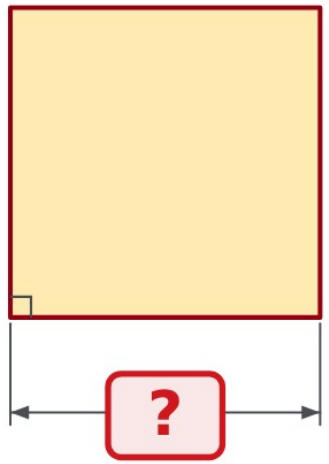
Convert 5/8 in to mm.

Answer independently. Keep worked solutions closed.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Independent check | 8C Q2

The square has an area of 3,136 mm². Find the side.



Answer independently. Keep worked solutions closed.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Wrap up and choose the next step

Identify what improved, what needs another attempt and which resource will support the next step.

Name the math and another essential skill used; record a concrete follow-up action.

Resources: Stations: 1B Q3 p.8; 4A Q14(a),15(a) p.26; 5A Q12–13 p.33; 6A Q19 p.40; 7A Q11–12 p.45; 8C Q1–2 p.55.

Textbook Chapter 1–8, selected; formula sheet p.219; conversion factors pp.220–221

Break

10:15 AM–10:30 AM

Return ready for: Review: geometry and job problems

This break is excluded from teaching minutes.

Review: geometry and job problems

Today’s goal: Select perimeter, area, volume or triangle methods from a drawing and report the correct unit.

Identify one target skill and the checking habit to use today.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Exercise Book 9A Q1,8 p.57; 10B Q1,11 pp.65–66; 11A Q7,19 pp.69–70; 12B Q3 p.77; 13A Q6,14 p.82.

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Recall and reconnect

Display three drawings and ask which result uses mm, mm^2 or mm^3; retrieve DEG mode for trig.

Review previous feedback and attempt the retrieval prompt without displayed solutions.

Resources: Exercise Book 9A Q1,8 p.57; 10B Q1,11 pp.65–66; 11A Q7,19 pp.69–70; 12B Q3 p.77; 13A Q6,14 p.82.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Match the drawing to the requested quantity

1 Outside boundary is perimeter: add edges, or use 2(l + w) for a rectangle. Report a length unit such as mm or m.

2 Surface coverage is area: use the appropriate area formula and subtract openings where stated. Report square units.

3 Space or capacity is volume: base area × height for a prism. Report the requested cubic or capacity unit.

4 A right-triangle side uses Pythagoras or the selected trig ratio. The side answer is a length; confirm DEG mode for trig.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Use the drawing to check the calculation

1. Identify the unknown before selecting the formula, and label all given dimensions on the appropriate drawing.

2. Keep each stated cut and kerf with its own problem; include the stated kerf in the paired work on 12B Q3.

3. Reject an answer whose unit does not match the quantity or whose magnitude does not fit the drawing.

4. For the scheduled fresh Hub question, explain why the formula applies and retain the question/code for checking.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Match the quantity, formula and unit

Find	Method example	Result unit
Outside boundary	Add edges; rectangle 2(l + w)	mm or m
Surface coverage	Area; subtract openings	mm^2 or m^2
Space / capacity	Base area × height	mm^3, m^3 or L
Right-triangle side	Pythagoras or selected trig ratio	mm or m

Together • Work 10B Q1 and 12B Q3 in pairs.
Annotate every dimension and include stated kerf. Explain why the method matches the unknown; reject an unreasonable unit or magnitude.

Exercise Book pp.65,77 • Shape Canvas rectangle, prism and right triangle • Formula sheet p.219

Textbook | page 172

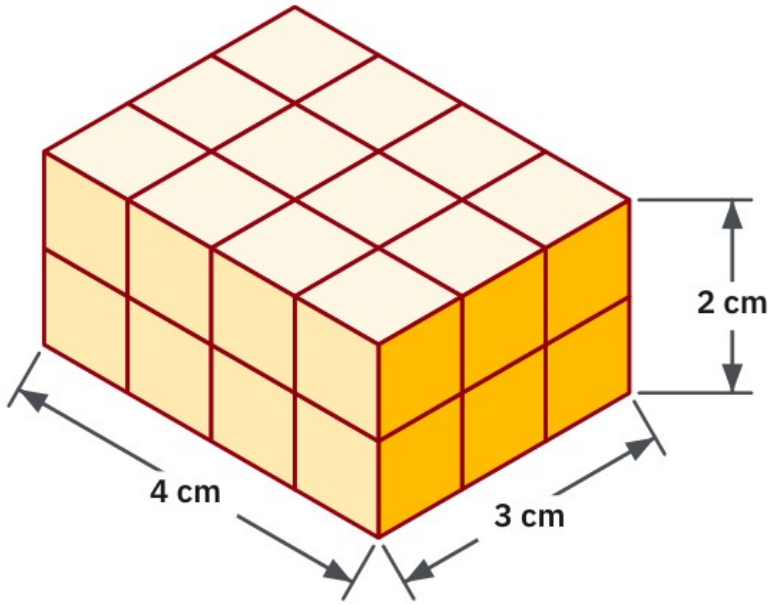


Figure 11.1 One layer holds 4 × 3 = 12 cubes of 1 cm³. Two layers hold 12 × 2 = 24 cm³.

Figure 11.1: volume counts unit cubes in three directions.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Textbook | page 172

Changing between cubic units uses the length factor three times. A cubic centimetre is 10 mm × 10 mm × 10 mm = 1,000 mm³. A cubic metre is 1,000 mm × 1,000 mm × 1,000 mm = 1,000,000,000 mm³. A cubic foot is 12 in × 12 in × 12 in = 1,728 in³.

THREE FACTORS, NOT ONE

To change mm³ to m³, divide by 1,000,000,000 (1,000 three times). Dividing by 1,000 only changes a length. Many people find it easier to change each length to metres first and then multiply.

Cubic conversions use the length factor three times.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Spot the mistake. Explain the correction.

A surface-coverage calculation is labelled mm³. What does that unit suggest, and what should you recheck?

Discuss the method before checking the answer.

Cubic units describe volume. Recheck the requested quantity and formula: surface coverage needs an area calculation and square units; a correct-looking number with the wrong unit is not a complete result.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Work together

Work Exercise Book 10B Q1 and 12B Q3 in pairs. Annotate all dimensions and explicitly include stated kerf.

Explain the method and check directions; work with a partner only where permitted.

Resources: Exercise Book 9A Q1,8 p.57; 10B Q1,11 pp.65–66; 11A Q7,19 pp.69–70; 12B Q3 p.77; 13A Q6,14 p.82.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Guided practice | 10B Q1

A mezzanine floor is 7.2 m by 3.6 m. It is covered with checker plate sheets that are 1.2 m by 2.4 m, laid edge to edge with no waste. Find the floor area and the number of sheets.

Work with a partner. Take turns explaining the next step.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Compare your method | 10B Q1

A mezzanine floor is 7.2 m by 3.6 m. It is covered with checker plate sheets that are 1.2 m by 2.4 m, laid edge to edge with no waste. Find the floor area and the number of sheets.

CHECK TOGETHER

25.92 m², 9 sheets

7.2 × 3.6 = 25.92 m², one sheet = 2.88 m², 25.92 ÷ 2.88 = 9

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Your scheduled practice

Solve the remaining listed exercises, prioritizing documented needs. Finish with one fresh Numeracy Hub question; explain why that formula applies and keep its question/code for checking.

Keep numbered written solutions and units. Use tools only under the conditions stated for this block.

Resources: Exercise Book 9A Q1,8 p.57; 10B Q1,11 pp.65–66; 11A Q7,19 pp.69–70; 12B Q3 p.77; 13A Q6,14 p.82.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Open Numeracy Hub

ESSENTIAL SKILLS FOR ABORIGINAL FUTURES · SKILLS FOR SUCCESS: NUMERACY

ACCESS Trades Numeracy Hub

Every worked example shows the calculator keys. Open the calculator any time with the button at the bottom right.

Calculator TI-30Xa

1 Shape Studio | 2 Order of Operations | 3 Volume and Layout | 4 Angles and Trig | 5 Fabrication | 6 Pipe Trades | 7 Unit Conversions | 8 Mechanical Aptitude | 9 Coordinate Systems | 10 Measuring Tools | 11 Exam builder

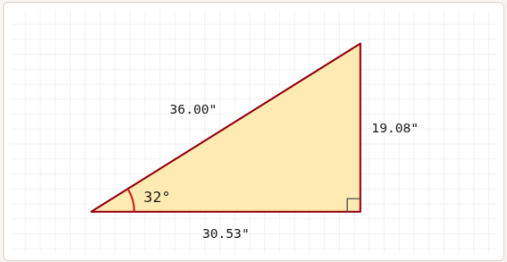
A– 100% A+ Hide operations

MODULE 3

Right-triangle trigonometry

Braces, gussets, ramps, cut angles and tapers are all right triangles. Trig finds the missing side or angle without laying it out full size.

SOH: sin = opposite ÷ hypotenuse. CAH: cos = adjacent ÷ hypotenuse. TOA: tan = opposite ÷ adjacent. Set your calculator to degrees (DEG).



Worked example

A brace 36.00" long leans at 32° from the floor. How high up the post does it reach, and how far out along the floor?

1. Label the sides from the 32° angle. The brace is the hypotenuse. The rise is opposite. The run is adjacent.
2. Rise uses SOH: opposite = hypotenuse × sin 32° = 36.00 × 0.5299 = 19.08".
3. Run uses CAH: adjacent = hypotenuse × cos 32° = 36.00 × 0.8480 = 30.53".
4. Check with Pythagoras: $19.08^2 + 30.53^2 \approx 36.00^2$.

Answer: Rise (opposite) 19.08 in, Run (adjacent) 30.53 in

TRY OTHER NUMBERS

Angle 32°

Brace length (hypotenuse) 36.00"

On the TI-30Xa Start with ON/C ON/C and DEG on the screen.

Rise

36 × 32 SIN =

SCREEN 19.07709351 Run on calculator

Calculator

Navigation example. Use the values and units in the assigned task.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Practise in Numeracy Hub and Shape Canvas

1. In Shape Canvas, compare the scheduled rectangle, rectangular prism and right triangle. Label the given dimensions and identify the unknown for the relevant written problem.

2. After the listed paper work, use Numeracy Hub Volume and Layout or Angles and Trig for one Easy task in a taught topic, chosen to match the documented need.

3. Keep the question/code, written formula, diagram and units. Explain why the method applies and use the drawing to reject an unreasonable unit or magnitude.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Practise from the Exercise Book | 9A Q1

Find the perimeter of a rectangle 2,400 mm by 1,150 mm.

Write the operation, working and units before checking.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Practise from the Exercise Book | 9A Q8

A rectangle has a perimeter of 3,800 mm and a length of 1,200 mm. Find the width.

Write the operation, working and units before checking.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Check independently

Exercise Book 10B Q11, 11A Q19 and 13A Q14, with diagrams and units.

Submit work and demonstrate the checking strategy; identify a remaining difficulty.

Resources: Exercise Book 9A Q1,8 p.57; 10B Q1,11 pp.65–66; 11A Q7,19 pp.69–70; 12B Q3 p.77; 13A Q6,14 p.82.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Independent check | 10B Q11

A 300 mm by 200 mm access hole is cut in a 1,200 mm by 800 mm cover plate. Find the area of plate left in mm² and in m².

Answer independently. Keep worked solutions closed.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Independent check | 11A Q19

A box holds 0.24 m³. Its base is 0.8 m by 0.5 m. Find its height.

Answer independently. Keep worked solutions closed.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Independent check | 13A Q14

A slope rises 120 mm over a run of 6,000 mm. Find the percent grade and the angle to two decimal places.

Answer independently. Keep worked solutions closed.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219

Wrap up and choose the next step

Identify what improved, what needs another attempt and which resource will support the next step.

Name the math and another essential skill used; record a concrete follow-up action.

Resources: Exercise Book 9A Q1,8 p.57; 10B Q1,11 pp.65–66; 11A Q7,19 pp.69–70; 12B Q3 p.77; 13A Q6,14 p.82.

Textbook Chapter 9–13 pp.140–214, selected; formula sheet p.219